

Duncan Yu

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Who Am I

I'm a Computing student at Queen's University who enjoys turning data into insight and impact. With strong programmatic thinking skills and a growing focus on Generative AI, I'm eager to contribute to part-time projects involving data cleaning, preprocessing, and AI development.

Qualifications

- **AI Engineering: GenAI & LLMs, Data Preparation, RAG, Fine-Tuning/Alignment, Multimodal AI/Agents**
 - **Extensive knowledge** in Game development, App Design, Data Modeling, Datawarehouse concept & framework
 - Self-discovery, but with guidance from professional, **aiming at exceeding expectations**
- AI:
- LLMs, Prompt Engineering, RAG, Supervised Fine-Tuning (LoRA, QLoRA), AI Agent Automation
 - Data Cleaning & Pretraining, pipelines, Web Scraping, OCR/ASR, synthetic data generation
 - Fine-tuning, SFT, LoRA, Deepspeed, TRL, **Cloud GPU computing** (50+ hrs)
 - Frameworks: Hugging Face, Transformer, LangChain, vLLM, Deepspeed, TRL, Gradio
 - Multimodal AI (Voice Agents, ASR/TTS), Agents & Workflow Orchestration (LangChain, MCP).
- Programming & Platforms:
- Programming languages: **Python** (3 years), PySpark (1 year), SQL (2 years), C# (2 years), VBA
 - Tools: Huggingface, PyTorch, TensorFlow, scikit-learn, FastAPI, Swagger, Postman, Claude
 - SQL Server, PostgreSQL, Snowflake, Databricks, AWS & Azure cloud, **Jira**, Confluence, DevOps

Working Experience

Generative AI Engineering Bootcamp | *Inference.ai* | 2025 (10 weeks)

- Gained hands-on experience with generative AI and large language models, focusing on prompt design, agent development, and Transformer architectures.
- Built and processed datasets through web scraping, OCR/ASR pipelines, and data cleaning.
- Applied retrieval-augmented generation (RAG) using embeddings and vector databases.
- Fine-tuned LLMs on GPU platforms using multiple methods, including FFT, LoRA, QLoRA and RLHF.
- Explored multimodal AI by developing voice agents and implementing real-time audio processing.

- Projects:**
- **Data preparation & SFT: Webscraping, data cleansing, de-dup, OCR, RAG**
 - **AI Agent Development: Function Calling, Model Context Protocol, Multi-Tool with Python**
 - **App: Student Expense AI Assistant**

Data Engineer Intern | *EDPYPF Services Incorporated* | 2022 – 2024 (2 years - part time)

- Developed Data Mart framework using Python and SQL to load data into PostgreSQL DB from multiple files
- Troubleshooting data gap issue in snowflake and databricks environment to identify and resolve them.
- Enhanced reconciliation process by delivering Python script to identify **DQ** issues - missing values and columns
- Built company website in 3 weeks, improving brand broadcasting and online presence.

- Projects:**
- **Datawarehouse framework and Pipeline development in Python and SQL**
 - **Automated Data Cleansing and Statistic Reporting in Excel**
 - **Company Website Development**

Lead Instructor & Python Curriculum Developer | *Envision Robotics* | 2023 – 2025 (2 years - Summer)

- Facilitated teams of volunteers to educate and engage younger generations in STEM activities in Summer Camps.
- Incorporate problem solving skills to resolve scheduling conflicts and issues that may arise in routine operations.
- Create an adaptable Python curriculum tailored to educating and engaging young minds.
- Use/develop various software to run machinery such as laser cutters, 3D printers and robots.

Education & Activities

- Computing** *Queen's University* 2025 | Waterloo Canadian Computing Competition **Top 5%**
- Activities:** Kickstarted and instructed 8 Globemaster's Coding Club, teaching cadets advanced Python skills (2 years)
Minecraft Modding with Java, Camping & Survival Instructor